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ASSEMBLY OF A RECEIVER WITH A BARREL

Abstract

An assembly of a receiver (**1**) with a barrel (**4**) wherein the barrel (**4**) is inserted into a barrel hole (**7**) in the receiver (**1**) and is attached in a replaceable manner by means of a clamping connection that comprises a longitudinal slot (**2**) that longitudinally divides the receiver (**1**) wall in the insertion location of the barrel (**4**). In the location of the longitudinal slot (**2**) at least two transversal screws (**3**) pass through the receiver (**1**). In the location of the longitudinal slot (**2**), a transversal hole (**5**) extends into the receiver (**1**) and a transversal pin (**10**) passes through the transversal hole (**5**) in such a way that the transversal pin (**10**) reaches into a corresponding transversal recess (**9**) in the barrel (**4**). Perpendicularly to the longitudinal axis of the transversal pin (**10**) a threaded hole (**11**) with a connecting screw (**12**) passes through the transversal pin (**10**). The connecting screw (**12**) further passes through a straight-through hole (**6**) in the receiver (**1**) and through a stock/frame (**13**) of the firearm to connect the receiver (**1**) to the stock/frame (**13**) of the firearm.

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Background/Summary

TECHNICAL FIELD

[0001] The invention relates to an assembly of a receiver with a barrel, the barrel being inserted into a barrel hole in the receiver and being attached in a replaceable manner by means of a clamping connection that comprises a longitudinal slot that longitudinally divides the wall of the receiver in the insertion location of the barrel, and at least two transversal screws passing through the receiver in the location of the longitudinal slot, and a transversal hole extending into the receiver In the location of the longitudinal slot, and a transversal pin passing through the transversal hole in such a way that the transversal pins reaches into a corresponding transversal recess in the barrel.

BACKGROUND ART

[0002] In non-modular firearms, i.e., without the possibility of replacing the barrel on the user level, the barrel is often directly screwed into the receiver. The barrel may also be connected to the receiver by pressing, or with the use of a pin.

[0003] Concerning modular firearms, their barrel can either be replaced without the use of tools or with the use of a minimal number of tools, preferably a single tool.

[0004] Firearms wherein the barrel can be removed without the use of tools are also called “take-down” firearms and have various barrel fixing mechanisms. Various lever cogs that are controlled from the outside or bayonet mechanisms are used. However, these types of firearms are not designed for easy replacement of the barrel. They provide the possibility of disassembling the firearm, mostly into two subgroups (barrel, the rest) to dramatically reduce the length, e.g., for transport.

[0005] In firearms where the barrel can be removed with the use of tools, these tools may be stored inside the gun, or are supplied together with the firearm as accessories. This type of firearms is not primarily intended for disassembly of the firearm for storage or transport, but their design does not make this impossible.

[0006] A known design of a firearm where the barrel can be removed using a tool features a barrel inserted into the receiver and attached in a replaceable manner using a clamping connection that comprises a longitudinal slot that longitudinally divides the wall of the receiver in the barrel insertion location having a slightly bigger length that the length of insertion of the barrel into the receiver. In the location of the longitudinal slot, at least two transversal screws pass through the receiver. Such a clamping connection then clamps to receiver in the region of the slot to fix the barrel.

[0007] Such a connection is very firm and stable and does not affect accuracy. In addition, it is relatively easy to produce, and thus cheap.

[0008] A disadvantage of such a design is that it makes it possible to mount the barrel in a different position or to leave the transversal screws poorly tightened. This may result in a slight extension/rotation of the barrel and consequently endangering of the shooter.

[0009] In the design according to the application US2013/0219765A1, the head of a screw reaches into a recess in the barrel. However, this design does not deal with the connection of the receiver and barrel with the frame, or the stock. In practice this means that the shooter's safety is only guaranteed in case the screws of the clamping connection are insufficiently or wrongly tightened. But this design does not cover the case these screws are completely missing in the firearm due to

the human factor. In such a case, the forestock can be quite normally screwed onto the firearm, but the barrel will not be safely attached/fixed.
[0010] The object of the invention is to propose such an improvement of a known assembly of the receiver with the barrel that will not have the above-mentioned shortcomings of the prior art.
SUMMARY OF THE INVENTION
[0011] The said goal is achieved through an assembly of a receiver with a barrel wherein the barrel is inserted into a barrel hole in the receiver and is attached in a replaceable manner by means of a clamping connection that comprises a longitudinal slot, that longitudinally divides the wall of the receiver in the barrel insertion location, and in the location of the longitudinal slot, at least two transversal screws pass through the receiver, and in the location of the longitudinal slot, a transversal hole extends into the receiver and a transversal pin passes through the transversal hole in such a way that the transversal pin reaches into a corresponding transversal recess in the barrel, according to the invention the principle of which is that a threaded hole with a connecting screw passes through the transversal pin perpendicularly to the longitudinal axis of the transversal pin wherein the connecting screw further passes through a straight-through hole in the receiver and through the stock/frame of the firearm to connect the receiver to the stock/frame of the firearm.
[0012] An advantage of the assembly according to the invention is that the threaded hole passing through the transversal pin perpendicularly to the longitudinal axis of the transversal pin with the connecting screw screwed in, which further passes through the straight-through hole in the receiver and through the stock/frame of the firearm, will eliminate the risk of inadvertent release of the barrel even in case the transversal screws in the receiver are completely absent due to the human factor, All this even if the user forgets to insert the transversal pin into the receiver because this will make the connection with the stock/frame of the firearm impossible and the user will immediately notice that the firearm is not properly assembled.
[0013] This is mainly important in modular rifles where the user can replace barrels of the firearm by him-/herself. The solution according to the invention provides the user with safety in case of any wrong replacement of the barrel.
[0014] In preferred embodiments, the transversal pin may have both a circular cross-section, and a non-circular cross-section.
[0015] In a preferred embodiment, the transversal hole comprises of a blind hole.
[0016] In another preferred embodiment, a spring-loaded catch protrudes from the surface of the transversal pin to snap into the corresponding recess in the receiver.
[0017] In still another preferred embodiment, a guiding groove is provided in the transversal pin surface that a stop attached to the stock/frame of the firearm extends into.

Description

BRIEF DESCRIPTION OF DRAWINGS
[0018] The invention will be described in a more detailed way with reference to particular embodiment examples of the assembly of the receiver with the barrel according to the invention, represented in the attached drawings where individual figures show:
[0019] FIG. **1**—vertical longitudinal section of the assembly of the receiver with the barrel according to the invention, attached to the firearm stock;
[0020] FIG. **2**—view of the receiver from the right side;
[0021] FIG. **3**—vertical cross-section of the receiver;
[0022] FIG. **4**—bottom view of the receiver;
[0023] FIG. **5**—view of the barrel from the right side;
[0024] FIG. **6**—axonometric view of the transversal pin;
[0025] FIG. **7**—an embodiment example with a transversal pin fitted with a spring-loaded catch

and a guiding groove with a stop wherein the transversal pin is completely withdrawn;
[0026] FIG. **8**—the embodiment of FIG. **7** wherein the transversal pin is completely inserted.
EXAMPLES OF EMBODIMENTS
[0027] The assembly of the receiver **1** with the barrel **4** is shown as attached to the stock **13** of a long firearm in FIG. **1**.
[0028] The barrel **4** is inserted into the barrel hole **7** in the receiver **1** and is attached in a replaceable manner by means of a clamping connection that comprises of a longitudinal slot **2** that longitudinally divides the wall of the receiver **1** in the Insertion location of the barrel **4**. The clamping connection further comprises of three transversal screws **3** that pass through the receiver **1** in the location of the longitudinal slot **2**.
[0029] In the location of the longitudinal slot **2**, a transversal hole **5** extends into the receiver **1**, the transversal hole being a blind hole and ending in the location of the slot **2** in the displayed example (see FIG. **3**). A transversal pin **10** passes through the transversal hole **5** in such a way that the transversal pin **10** reaches into the corresponding transversal recess **9** in the barrel **4**.
[0030] In FIG. **3**, an intersection **8** of the transversal hole **5** and the opening **7** for the barrel **4** in the receiver **1** is perceivable.
[0031] A threaded hole **11** passes through the transversal pin **10** perpendicularly to the longitudinal axis of the transversal pin **10** (see FIGS. **1** a **6**), a connecting screw **12** being screwed into the threaded hole, passing at the same time through the straight-through hole **6** in the receiver **1** and through the stock/frame **13** of the firearm to connect the receiver **1** to the stock/frame **13** of the firearm (see FIG. **1**).
[0032] The fact whether the receiver **1** is connected to the stock **13** or frame **13** of the firearm depends on whether the assembly according to the invention is applied to a long or short firearm.
[0033] In the embodiment examples shown, the transversal pin **10** has a circular cross section, but skilled persons will appreciate that the transversal pin **10** may also have a non-circular cross-section, e.g., a rectangular one.
[0034] FIGS. **7** and **8** show another embodiment example of the assembly according to the invention where a spring-loaded catch **14** protrudes from the surface of the transversal pin **10** to snap into a corresponding recess in the receiver **1**. The spring-loaded catch **14** is a ball that is pushed by a not shown spring in such a way that it partly protrudes over the surface of the transversal pin **10**. During the insertion of the transversal pin **10** into the transversal hole **5**, the ball is first pushed into the transversal pin **10** against the spring pressure, and subsequently it snaps into a recess in the receiver **1**, which is not shown here.
[0035] The end positions of the inserted and withdrawn transversal pin **10** are delimited by a stop **15** that is fixed in the stock **13** of the firearm and extends into the guiding groove **16** that is created in the transversal pin **10**.
[0036] During the replacement of the barrel **4**, both to change the calibre, or just to replace a worn barrel **4**, no intervention of a specialist gunmaker is necessary.
[0037] All that the user needs to do is to release the transversal screws **3** and withdraw the transversal pin **10** so that it should not extend into the transversal recess **9** in the barrel **4**, which will cause releasing of the barrel, so the barrel **4** will get free and it can be freely withdrawn from the opening **7** in the receiver **1**. After inserting of the new barrel **4**, this barrel **4** will then be fixed in an analogous manner.
LIST OF REFERENCE SIGNS
[0038] **1** receiver
[0039] **2** slot
[0040] **3** transversal screw
[0041] **4** barrel
[0042] **5** transversal hole
[0043] **6** straight-through hole

[0044] **7** barrel hole
[0045] **8** intersection of the transversal hole and the barrel hole
[0046] **9** transversal recess
[0047] **10** transversal pin
[0048] **11** threaded hole
[0049] **12** connecting screw
[0050] **13** stock/frame of the firearm
[0051] **14** spring-loaded catch
[0052] **15** stop
[0053] **16** guiding groove

Claims

1. An assembly of a receiver (**1**) with a barrel (**4**) wherein the barrel (**4**) Is inserted into a barrel hole (**7**) in the receiver (**1**) and is attached in a replaceable manner by means of a clamping connection that comprises a longitudinal slot (**2**) that longitudinally divides the receiver (**1**) wall in the insertion location of the barrel (**4**) and in the location of the longitudinal slot (**2**), at least two transversal screws (**3**) pass through the receiver (**1**), and in the location of the longitudinal slot (**2**), a transversal hole (**5**) extends into the receiver (**1**) and a transversal pin (**10**) passes through the transversal hole (**5**) in such a way that the transversal pin (**10**) reaches into a corresponding transversal recess (**9**) in the barrel (**4**), characterized in that perpendicularly to the longitudinal axis of the transversal pin (**10**) a threaded hole (**11**) with a connecting screw (**12**) passes through the transversal pin (**10**) wherein the connecting screw (**12**) further passes through a straight-through hole (**6**) in the receiver (**1**) and through a stock/frame (**13**) of the firearm to connect the receiver (**1**) to the stock/frame (**13**) of the firearm.
2. The assembly according to claim 1, characterized in that the transversal pin (**10**) has a circular cross-section.
3. The assembly according to claim 2, characterized in that the transversal pin (**10**) has a non-circular cross-section.
4. The assembly according to claim 1, characterized in that the transversal hole (**5**) comprises of a blind hole.
5. The assembly according to claim 1 **4**, characterized in that a spring-loaded catch (**14**) protrudes from the surface of the transversal pin (**10**) to snap into a corresponding recess in the receiver (**1**).
6. The assembly according to claim 1, characterized in that in the surface of the transversal pin (**10**), a guiding groove (**16**) is created that a stop (**15**) that is attached to the stock/frame (**13**) of the firearm reaches into.